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CHE 210 - Homework set #11

(1) Murphy 6.47

Acetylene $\overset{\text{must be}}{>}$ expensive than methane

assume fuel is mixed w/air @ 25°C, and reacts adiabatically

t_{out} est flame temp

1) 25
2) 25
3) 15
4) 25

$\frac{90}{100}$

$$2\, C_2H_2 + 5\, O_2 + N_2 \longrightarrow 4\, CO_2 + 2\, H_2O + N_2$$

$$5\, O_2 \left(\frac{79\, g\, N_2}{21\, g\, O_2}\right) = 18.8 \text{ mol } N_2$$

$$\Delta H = Cp_{i,avg}\, \Delta T \Rightarrow \sum_P n_e \hat{H}_e = \Delta T \left[4\, (Cp)\, CO_2 + 2\, (Cp)\, H_2O + 18.8 (Cp)\, N_2\right]$$

$$\sum_P n_e \hat{H}_e = (T_f - 298K)\left[4(45 \tfrac{kJ}{mol}) + 2(35 \tfrac{kJ}{mol}) + 18.8\,(30 \tfrac{kJ}{mol})\right]$$

$$= (T_f - 298K)(814 \tfrac{kJ}{K})$$

$$\hat{H}_f = 4 \text{ kmol} (-393{,}522 \tfrac{kJ}{kmol}) + 2 \text{ kmol} (-241{,}827 \tfrac{kJ}{kmol}) - 2 \text{ kmol} (226{,}700 \tfrac{kJ}{kmol})$$

$$= 2.5 \times 10^6 \text{ kJ}$$

$$(T_f - 298K)(814 \tfrac{kJ}{K}) = 2.5 \times 10^6 \text{ kJ}$$

$$T_f = \underline{3382.9 \text{ K}} \quad \text{acetylene}$$

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$$CH_4 + 2\, O_2 + 7.5\, N_2 \longrightarrow CO_2 + 2\, H_2O + 7.5\, N_2$$

$$\hat{H}_f = 1 \text{ kmol} (-393.522 \text{ kJ/kmol}) + 2 \text{ kmol} (-241{,}827 \tfrac{kJ}{kmol}) - 1 \text{ kmol} (-74{,}444 \tfrac{kJ}{kmol})$$

$$= -802{,}732 \text{ kJ}$$

$$\sum n_e \hat{H}_e = 802{,}732 \text{ kJ} = (T_f - 298K)\left[1(45 \tfrac{kJ}{mol}) + 2(35 \tfrac{kJ}{mol}) + 7.5\,(30 \tfrac{kJ}{mol})\right]$$

$$802{,}732 \text{ kJ} = (T_f - 298K)(340 \tfrac{kJ}{K})$$

$$(T_f - 298K)(340 \tfrac{kJ}{K}) = 802{,}732 \text{ kJ}$$

$$T_f = \underline{2658 \text{ K}} \quad \text{methane}$$

Acetylene is chosen because it has a higher T_f than methane (it is essentially hotter)

2. Murphy 6.48

40 mol % C_2H_6
19 mol % N_2
1 mol % H_2S
10 mol % O_2
30 mol % CO

waste gas stream
fed at 25°C
air fed at 25°C

→ flue gas exits furnace at 150°C

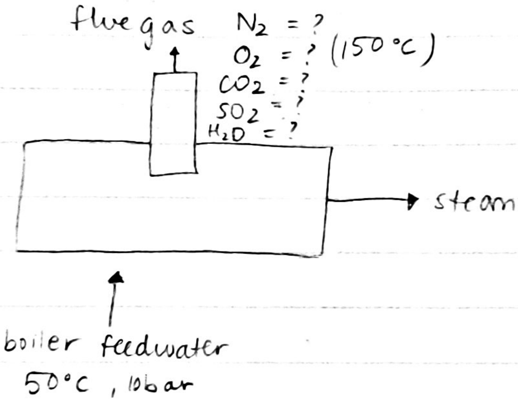
"complete combustion" ↳ assumed

generates sat. steam @10 bar, (boiler feedwater) 50°C and 10 bar

steam = ? [=] $\frac{\text{kg steam}}{\text{gmol gas}}$

waste gas stream (in)
$C_2H_6 = 0.40$
$N_2 = 0.19$
$H_2S = 0.01$
$O_2 = 0.10$
$CO = 0.30$

air @ 25°C
$O_2 = 0.21$
$N_2 = 0.79$



Basis: in Waste gas stream = 100 gmol/s

$n_{C_2H_6, waste} = 40$ gmol/s
$n_{N_2, waste} = 19$ gmol/s
$n_{H_2S, waste} = 1$ gmol/s
$n_{O_2, waste} = 10$ gmol/s
$n_{CO, waste} = 30$ gmol/s ✓

$C_2H_6 + \frac{7}{2}O_2 \rightarrow 2CO_2 + 3H_2O$ $\quad \xi_1$

$CO + \frac{1}{2}O_2 \rightarrow CO_2$ $\quad \xi_2$

$H_2S + \frac{3}{2}O_2 \rightarrow SO_2 + H_2O$ $\quad \xi_3$

$(\frac{7}{2})40 + (\frac{1}{2})30 + (\frac{3}{2})1 - 10 = 146.5$ gmol/s

Air - 20% excess

$n_{O_2, air} = (1.2)(146.5 \text{ gmol/s}) = 175.8 \text{ gmol/s}$

$n_{N_2, air} = (\frac{79}{21})(175.8 \text{ gmol/s}) = 661.3 \text{ gmol/s}$

Material balance

$n_{C_2H_6, flue} = 0 = 40 - \xi_1$ $\qquad \xi_1 = 40$ gmol/s

$n_{CO, flue} = 0 = 30 - \xi_2$ $\qquad \xi_2 = 30$ gmol/s

$n_{H_2S, flue} = 0 = 1 - \xi_3$ $\qquad \xi_3 = 1$ gmol/s

$n_{CO_2, flue} = 2\xi_1 + \xi_2 = 2(40) + 30 = 110$ gmol/s

$n_{SO_2, flue} = \xi_3 = 1$ gmol/s

$n_{H_2O, flue} = 3\xi_1 + \xi_3 = 3(40) + 1 = 121$ gmol/s

$n_{N_2, flue} = n_{N_2, waste} + n_{N_2, air} = (19 \frac{gmol}{s}) + 661.3 \frac{gmol}{s} = 680.3 \frac{gmol}{s}$

$n_{O_2, flue} = n_{O_2, waste} + n_{O_2, air} - \frac{7}{2}\xi_1 - \frac{1}{2}\xi_2 - \frac{3}{2}\xi_3$

$= [10 \text{ gmol/s} + 175.8 \text{ gmol/s} - \frac{7}{2}(40 \text{ gmol/s}) - \frac{1}{2}(30 \text{ gmol/s}) - \frac{3}{2}(1 \text{ gmol/s})]$

$= 29.3$ gmol/s

$$H_{flue} = \sum_K \xi_K \Delta \hat{H}_{c,K} + \sum_i n_{i, flue} \int_{T_{in}}^{T_{out}} C_{p,i}\, dT$$

C_2H_6: $\Delta H^\circ_c = -1428.6 \frac{kJ}{gmol}$

CO: $\Delta H^\circ_c = -283 \frac{kJ}{gmol}$

H_2S: $\Delta H^\circ_c = -518.7 \frac{kJ}{gmol}$

CO_2: $\int_{298K}^{423K} 19.80 + 0.07344T - (5.602\times10^{-5})T^2 + 1.7115\times10^{-8}T^3\, dT$

$= 4968.68 \frac{J}{mol} \longrightarrow 4.97 \frac{kJ}{mol}$

H_2O: $\int_{298K}^{423K} 32.24 + 0.001924T + 1.055\times10^{-5}T^2 - 3.569\times10^{-9}T^3\, dT$

$= 4268.27 \frac{J}{mol} \longrightarrow 4.27 \frac{kJ}{mol}$

O_2: $\int_{298K}^{423K} 29.1 + 0.01158T - 6.076\times10^{-6}T^2 + 1.311\times10^{-8}T^3\, dT$

$= 4138.71 \frac{J}{mol} \longrightarrow 4.14 \frac{kJ}{mol}$

N_2: $\int_{298K}^{423K} 31.15 - 0.01357T + 2.680\times10^{-5}T^2 - 1.168\times10^{-8}T^3\,dT$

$= 3651.52 \frac{J}{mol} \longrightarrow 3.65 \frac{kJ}{mol}$

SO_2: $\int_{298K}^{423K} 23.85 + 0.06699T - 4.961\times10^{-5}T^2 + 1.328\times10^{-8}T^3\,dT$

$= 5185.99$ J/mol $\longrightarrow 5.18$ kJ/mol

$H_{flue} = [40(-1428.6 \text{ kJ/gmol}) + 30(-283 \frac{kJ}{gmol}) + 1(-518.7 \frac{kJ}{gmol})]$
$+ [110(4.97 \frac{kJ}{mol}) + 121(4.27 \frac{kJ}{mol}) + 29.3(4.14 \frac{kJ}{mol})$
$+ 680.3(3.65 \frac{kJ}{mol}) + 1(5.18 \frac{kJ}{mol})]$

$H_{flue} = -62480.8$ kJ/s

$H_{out} - H_{in} = Q \longrightarrow Q = -62480.8$ kJ/s

$m_{steam}(\hat{H}_{steam} - \hat{H}_{\substack{\text{boiling}\\\text{feedwater}}}) = m_{steam}(277.1 - 210.19) = -62480.8$ kJ/s

$m_{steam} = 24.3$ kg/s

Basis was 100 gmol/s so 0.243 $\frac{\text{kg steam}}{\text{gmol waste}}$ was burned

✓

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3 fuel gas 85 mol% methane, 15 mol% ethane } burned completely w/ oxygen @ 25°C. prods brought to back down to 25°C

(a) reactor is continuous

basis 1 mol/s

heat rate transferred from reactor per mole of fuel fed to the reactor

$\hat{Q} = \dot{Q}/\dot{n}$ [=](KJ/mol)

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

$C_2H_6 + \frac{7}{2}O_2 \rightarrow 2CO_2 + 3H_2O$

$(0.85 \text{ mol/s MeOH})(2 \text{ mol } O_2) = 1.7 \frac{mol}{s} O_2$

$(0.15 \text{ mol/s EtOH})(\frac{7}{2} \text{ mol } O_2) = 0.53 \frac{mol}{s} O_2$

$\Delta H_{f,prod} - \Delta H_{f,react} = \Delta H_r$

MeOH: $[(-393,522 \frac{J}{mol}) + 2(-241,825 \frac{J}{mol})] - [(-74,448 \frac{J}{mol})] = -802,728$ J/mol

EtOH: $[2(-393,522 \text{ J/mol}) + 3(-241,827 \text{ J/mol})] - [(-84,670 \text{ J/mol})] = -1427,855$ J/mol

$(0.85)(-802,728 \text{ J/mol}) + (0.15)(-1427855 \text{ J/mol}) = -896,497 \text{ J/s} \cdot \frac{s}{mol}$

$= 896.5 \frac{kJ}{mol}$ ✓

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(b) constant-volume batch reactor. heat transfer ~~from the reactor per mole~~ per mole of fueled charge / fuel fed (KJ/mol)

basis of 1 mol of fuel

↳ Batch → Internal E.

$\Delta H_{r,MeOH}(0.85) - \Delta H_{r,EtOH}(0.15)$

$(0.85)(-802,728 \frac{J}{mol}) - (0.15)(-1427855 \text{ J/mol}) = 896.5$ kJ/mol

(c) The parts in (a) and (b) do not depend on the percent excess oxygen because the reaction is not affected. (when oxygen is used in excess) α

$$H_3 - H_2 = \sum_i n_{i,2} \int_{275°C}^{25°C} C_{P,i}\, dT + \sum_e \Delta \hat{H}^\circ_{re} + \sum_i n_{i,3} \int_{25°C}^{T_3} C_{P,i}\, dT = 0$$

Solver from excel was used to obtain T_3 → Excel sheet?

$T_3 = 1300\ K$ which is the reactor outlet temp

↳ 1020°C

$$H_2 - H_1 = H_3 - H_4$$

$$\sum_i n_{i,2} \int_{25°C}^{275°C} C_{P,i}\, dT = \sum_i n_{i,3} \int_{T_4}^{1000} C_{P,i}\, dT$$

Solver from excel was used to obtain T_4

$T_4 = 1060K$ which is the product outlet temp

↳ 790°C

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